

ADVANCED MACHINE LEARNING · WORKSHEET 08

Polynomial Regression & Assumptions of Linear Regression

Feature Transformation · Overfitting · Five Assumptions

TEACHER COPY — Lecture 8

PART I: POLYNOMIAL REGRESSION

SECTION 1 Start from the Failure of a Straight Line

▷ HOOK

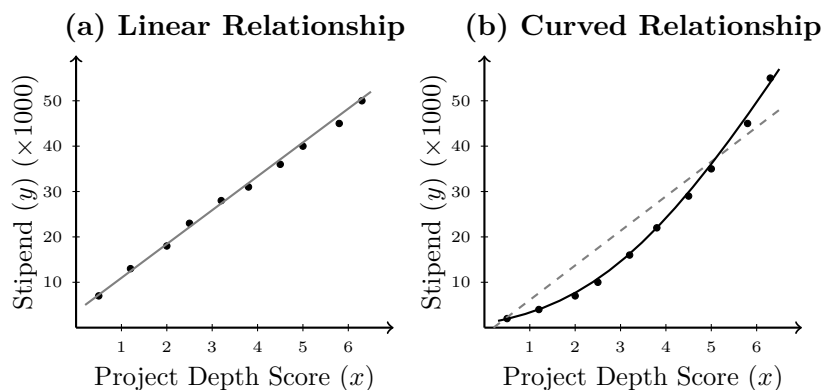
Suppose we want to predict the **internship stipend** of an NST student using one feature: a **project depth score** (1 = basic project, 10 = strong, production-like portfolio).

At first, higher project depth should produce higher stipend. But the relationship may not rise at a constant rate.

- Moving from score 2 to 4 may help a little.
- Moving from score 8 to 10 may help *much more*, because strong projects unlock better companies and referrals.

The scatter plot may **curve upward**.

Visual: Linear vs. Curved Scatter



The ordinary one-feature Linear Regression model is:

$$y = \beta_0 + \beta_1 x$$

This has only **one slope**. The slope is constant. If x increases by one unit, the predicted value changes by the same amount everywhere.

But in the scatter plot, the effect of one more unit of project depth seems larger at high values of x than at low values. We need a model whose **slope can change** as x changes.

SECTION 2 The Simple Idea: Create a New Feature x^2

▷ HOOK

Do not think of Polynomial Regression as a completely new algorithm. Think of it as a **feature transformation**: creating a new input column from an existing input column.

For example:

- If the original feature value is $x = 4$, the new feature is $x^2 = 16$.
- If $x = 8$, then $x^2 = 64$.

Notice that x^2 grows faster when x is large. This gives the model a way to **bend upward**.

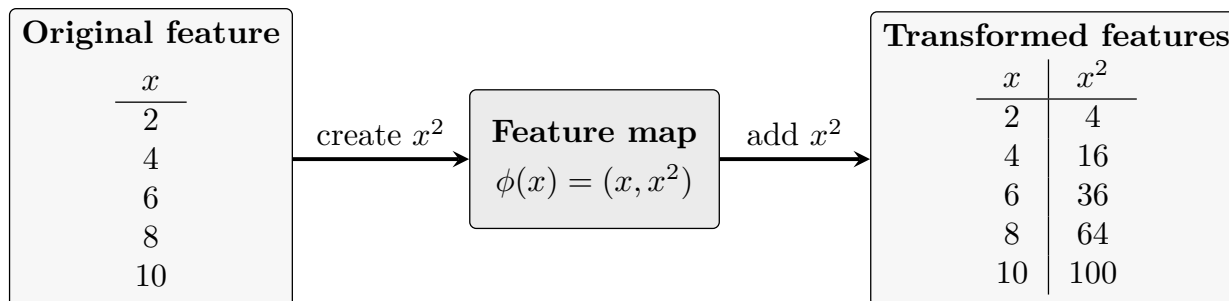
Example Dataset — NST Internship Stipend

Student	Raw feature x	New feature x^2	Target y (stipend in $\times 1000$)
A	2	<u>4</u>	24
B	4	<u>16</u>	42
C	6	<u>36</u>	62
D	8	<u>64</u>	91
E	10	<u>100</u>	128

After adding this new feature, the model becomes:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

Visual: One Input Column Becomes Two Model Features



★ KEY INSIGHT

A **feature space** is the coordinate system formed by the features we give to the model.

- With only one feature x , every student is located by **one** coordinate.
- With two features x and x^2 , every student is located by **two** coordinates.

The target y then becomes the vertical axis.

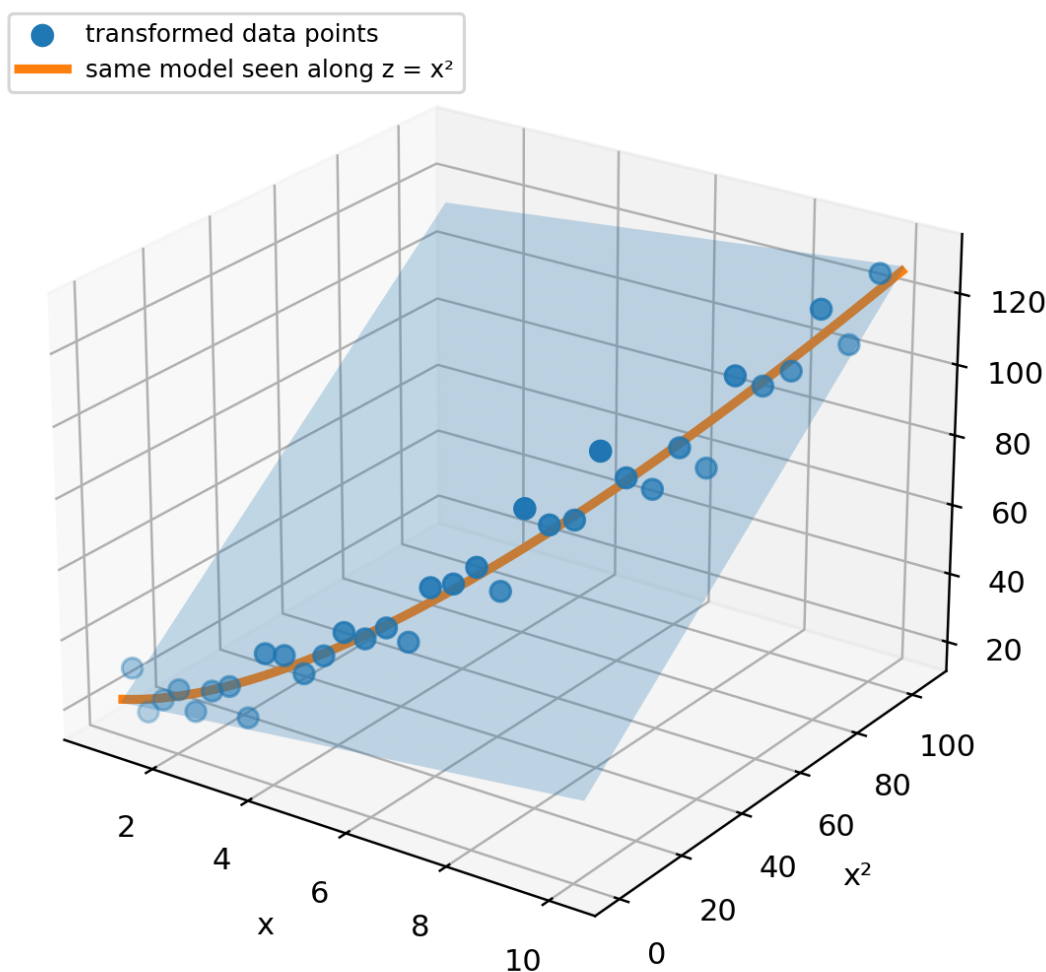
SECTION 3 The 3D Mental Model: A Curve Becomes a Plane

▷ HOOK

Let us rename x^2 as z for one minute: $z = x^2$. Then the model becomes $y = \beta_0 + \beta_1 x + \beta_2 z$. This is the equation of a **plane** in three-dimensional space with axes x , z , and y .

Visual: From 2D curve to 3D plane

In feature space, a quadratic model is a plane



In the original graph, we saw a curve. But after adding x^2 , the model is fitting a **plane**. When the fitted plane is viewed only along the path where $z = x^2$, it appears as a curve in the original (x, y) plot.

◆ TAKEAWAY

Polynomial Regression is not a new algorithm. We gave **Linear Regression** a **better coordinate system**. The model is still linear in its parameters $\beta_0, \beta_1, \beta_2$. It is only non-linear in the original feature x .

SECTION 4 Generalizing to Higher-Degree Polynomial Features

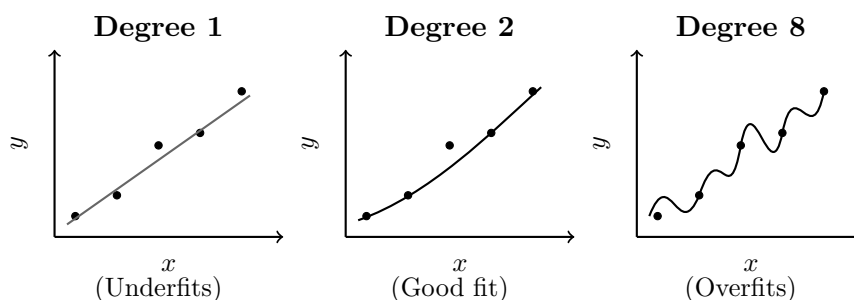
A **polynomial degree** is the highest power of x used in the model:

- Degree 1: $y = \beta_0 + \beta_1 x$ (straight line)
- Degree 2: $y = \beta_0 + \beta_1 x + \beta_2 x^2$ (can create a U-shape or upward bend)
- Degree 3: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$ (can bend more flexibly)

The general degree- n polynomial regression model is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_n x^n$$

Visual: Increasing degree and the risk of overfitting



△ WARNING — OVERFITTING

Overfitting means the model captures accidental fluctuations in the training data instead of the stable pattern that will repeat on future data. A very high-degree polynomial can twist itself to reduce training error, but that does not guarantee better prediction on new students.

Classroom question: If a degree-8 curve passes closer to every training point than a degree-2 curve, should we automatically prefer degree 8?

Expected answer: No. Lower training error is not enough. The curve must represent the **underlying relationship**, not the **noise in one dataset**.

PRACTICE P1 · POLYNOMIAL REGRESSION

- (a) Go back to the Example Dataset table in Section 2. Fill in the x^2 column for all five students.

Answer: $2^2 = 4$, $4^2 = 16$, $6^2 = 36$, $8^2 = 64$, $10^2 = 100$.

- (b) A degree-2 polynomial has the form $y = \beta_0 + \beta_1x + \beta_2x^2$. How many feature columns does the design matrix $\mathbf{X}$ have (including the intercept column of ones)?

3 columns ($1, x, x^2$)

- (c) Complete the table:

Degree	No. of feature columns (incl. intercept)	Can the curve bend?
1	<u>2</u>	<u>No (straight line)</u>
2	<u>3</u>	<u>Yes (one bend)</u>
5	<u>6</u>	<u>Yes (up to 4 bends)</u>
n	<u>$n + 1$</u>	<u>Yes (up to $n - 1$ bends)</u>

- (d) **True or False:** Polynomial Regression uses a different optimization algorithm than Linear Regression. [**F**]

Justify: It uses the same OLS or GD. We only changed the features, not the algorithm.

- (e) **Assertion–Reason:**

Assertion (A): A degree-20 polynomial on 25 data points will likely have lower training error than a degree-2 polynomial.

Reason (R): A higher-degree polynomial has more parameters and can twist itself to pass closer to every training point.

- (i) Both A and R are true, and R is the correct explanation of A. [☒]
- (ii) Both A and R are true, but R is not the correct explanation of A. [☐]
- (iii) A is true but R is false. [☐]
- (iv) A is false but R is true. [☐]

PART II: THE FIVE ASSUMPTIONS OF LINEAR REGRESSION

WHY ASSUMPTIONS MATTER

Polynomial Regression showed us an important lesson: Linear Regression is powerful when the representation of the data matches the real pattern. But after choosing a model, we still need to ask whether the model is **behaving in a trustworthy way**.

Linear Regression relies on several assumptions to ensure the validity and reliability of the estimates and inferences. Here, **inference** means understanding the relationship between variables, interpreting coefficients, and making statements about how changes in inputs affect the output.

Notation:

- **Residual** = actual value – predicted value: $e = y - \hat{y}$

The five key assumptions:

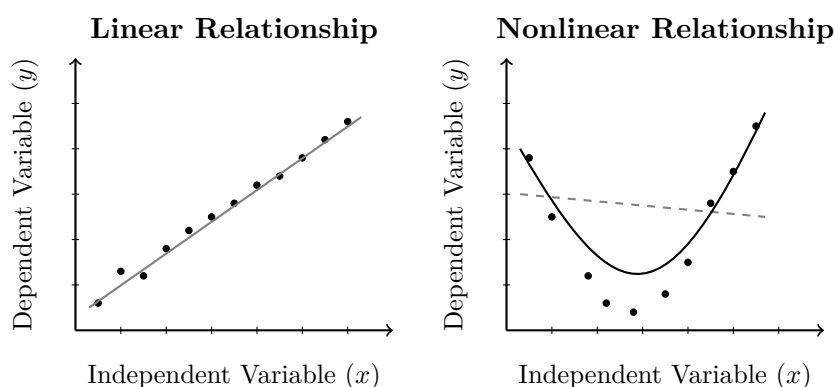
1. Linearity
2. Normality of Residuals
3. Homoscedasticity
4. No Autocorrelation of Errors
5. No (or little) Multicollinearity

SECTION 5 Assumption 1 — Linearity

▷ HOOK

The assumption: There is a **linear relationship** between the independent variables and the dependent variable. The model assumes that changes in the independent variables lead to proportional changes in the dependent variable.

Visual: Linearity satisfied vs. violated

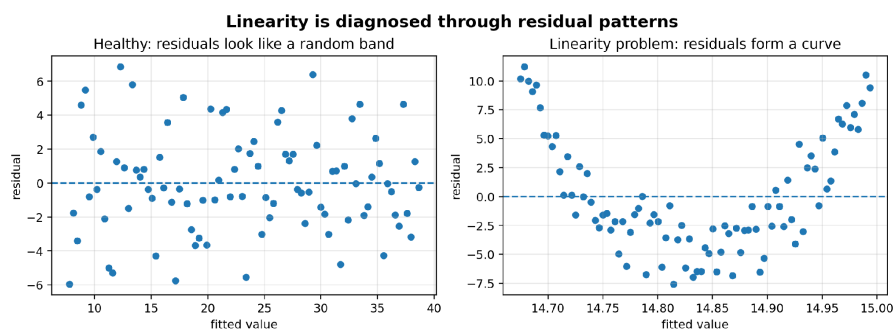


*Linearity means the central trend should be well represented by a straight line.
A curved pattern suggests violation of the linearity assumption.*

Linearity does **not** mean every point must lie exactly on a line. It means the *central trend* should be reasonably represented by a straight line or by a linear combination of features.

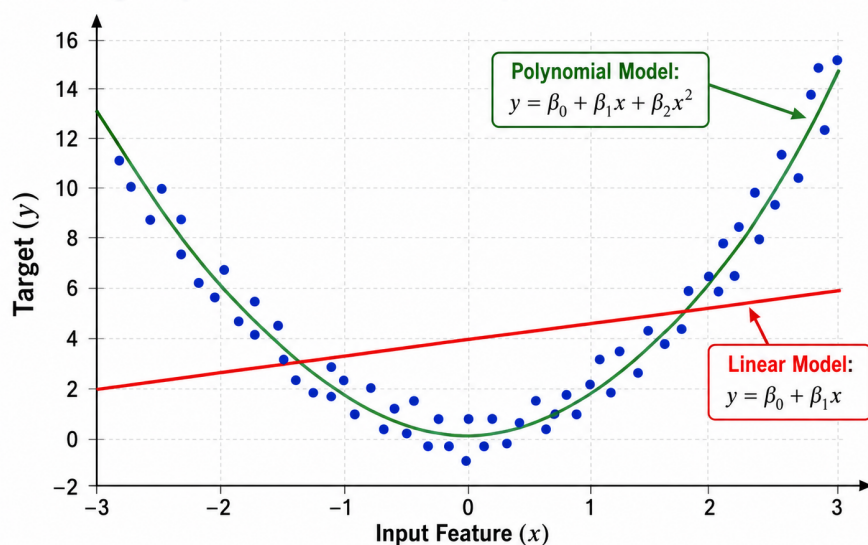
How to check this assumption:

1. **Scatter plots:** Create scatter plots of y against each independent variable.
2. **Residual plots:** Plot residuals against predicted values. Residuals should be **randomly scattered around zero**.



3. **Polynomial terms:** Add polynomial terms and compare model fit. If the new model significantly improves, linearity may be violated.

Adding Polynomial Terms Can Capture a Curved Relationship



If polynomial terms significantly improve the fit, the original linearity assumption may be violated.

What to do when the assumption fails:

- **Transformations:** Apply logarithmic, square-root, or other transformations.
- **Polynomial Regression:** Add polynomial terms to capture non-linear relationships.

SECTION 6 Assumption 2 — Normality of Residuals

▷ HOOK

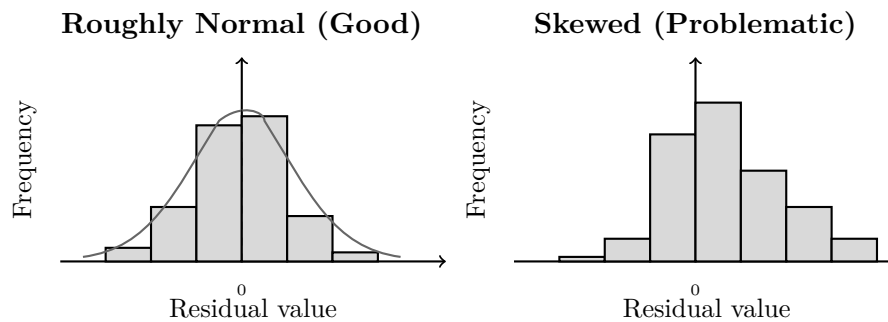
The assumption: The error terms (residuals) are assumed to roughly follow a **normal distribution** with mean zero and variance σ^2 .

Mean zero means the residuals are centered around zero.

Normal distribution means most residuals are small, and very large positive or negative residuals are rare.

How to check:

Histogram of residuals — plot a histogram and check whether it resembles a bell-shaped curve centered around zero.



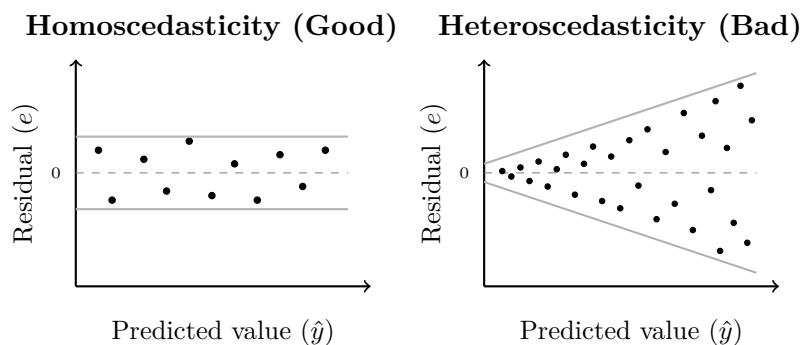
We are not expecting a perfect bell curve. We are checking whether the residuals are **roughly symmetric around zero** and whether extreme errors are not happening too often.

SECTION 7 Assumption 3 — Homoscedasticity

▷ HOOK

The assumption: The spread of the error terms (residuals) should be **constant** across all levels of the independent variables. If the spread changes systematically, it leads to **heteroscedasticity**.

How to check: Create a scatter plot of residuals against predicted values.



What to do when the assumption fails:

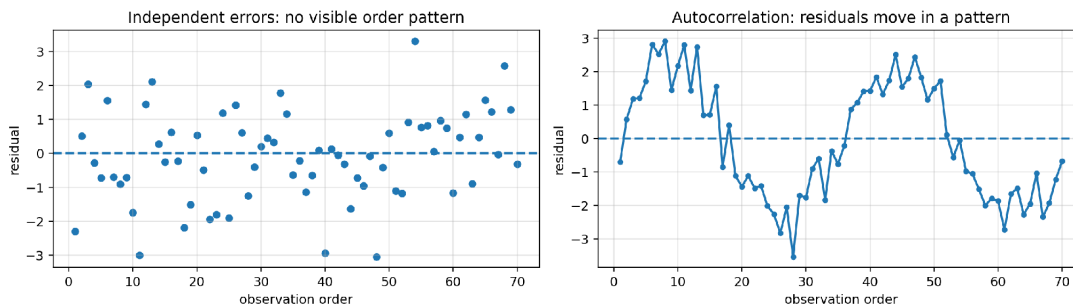
- **Transformations:** Apply logarithmic, square-root, or inverse transformations to stabilize the variance.

SECTION 8 Assumption 4 — No Autocorrelation of Errors

▷ HOOK

The assumption: There should be no apparent correlation or pattern in the residuals. **Independence of errors** means one residual should not help us predict the next residual. This is especially important when data has a natural order (time series).

Visual: Healthy random residuals vs. autocorrelation pattern

No autocorrelation: residuals should not follow an ordered pattern

If residuals move in a pattern, the model has left some structure in the data unexplained.

PRACTICE P2 · IDENTIFY THE ASSUMPTION

For each residual plot described below, identify **which assumption** is violated.

	Residual behaviour	Violated assumption
(a)	Residuals form a clear U-shaped curve when plotted against $\hat{y}$.	<u>Linearity</u>
(b)	Residual spread grows wider as $\hat{y}$ increases (funnel shape).	<u>Homoscedasticity</u>
(c)	Histogram of residuals is heavily skewed to the right.	<u>Normality</u>
(d)	Residuals in a time-series model show long runs of positive values followed by long runs of negative values.	<u>Autocorrelation</u>

SECTION 9 Assumption 5 — No Multicollinearity**▷ HOOK**

Multicollinearity occurs when two or more independent variables in a regression model are **highly correlated**. They exhibit a strong linear relationship, making it difficult to isolate the individual effect of each variable on the dependent variable.

9.1 The Setup

Suppose we have two features X_1 and X_2 , and we want to predict Y :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon_1$$

Here, ε_1 represents the part of Y not explained by the model.

If X_1 and X_2 are highly correlated, then:

$$X_1 = a_0 + a_1 X_2 + \varepsilon_2$$

If we substitute X_1 into the equation for Y , the model becomes:

$$Y \approx \underline{C_0 + C_1 X_2 + \varepsilon_3}$$

So, effectively, Y becomes a linear function of X_2 only.

9.2 Inference vs. Prediction

★ KEY INSIGHT

- If the goal is **inference**, multicollinearity is **bad** because β_1 and β_2 become unreliable.
- If the goal is **prediction only**, multicollinearity is usually **less harmful**. The model may still predict well, but the individual coefficient values may not be trustworthy.

9.3 Perfect Multicollinearity

If X_1 and X_2 are *perfectly* correlated:

$$X_1 = a_0 + a_1 X_2 \quad (\text{no error term})$$

Recall the OLS equation:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

When columns of $\mathbf{X}$ are a perfect linear combination, $\mathbf{X}^\top \mathbf{X}$ becomes **singular**.

9.4 Proof that $\mathbf{X}^\top \mathbf{X}$ Is Singular

Step 1: Suppose the columns of $\mathbf{X}$ are linearly dependent. There exists a non-zero vector $\mathbf{c}$ such that:

$$\mathbf{X} \mathbf{c} = \underline{\mathbf{0}}$$

Step 2: Multiply both sides by $\mathbf{X}^\top$:

$$\mathbf{X}^\top \mathbf{X} \mathbf{c} = \underline{\mathbf{X}^\top \mathbf{0}}$$

Step 3: Therefore:

$$\mathbf{X}^\top \mathbf{X} \mathbf{c} = \underline{\mathbf{0}}$$

Step 4: But $\mathbf{c} \neq \mathbf{0}$. So $\mathbf{X}^\top \mathbf{X}$ sends a non-zero vector to zero.

Therefore, $\mathbf{X}^\top \mathbf{X}$ has dependent columns.

Step 5: So $\mathbf{X}^\top \mathbf{X}$ is singular.

Step 6: Therefore: $\det(\mathbf{X}^\top \mathbf{X}) = \underline{\mathbf{0}}$

Step 7: Hence: $(\mathbf{X}^\top \mathbf{X})^{-1}$ does not exist

9.5 Numerical Example: 3×3 Matrix

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

Here, Column 3 = Column 1 + Column 2. The columns are linearly dependent.

Compute $\mathbf{X}^\top \mathbf{X}$:

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} \underline{\mathbf{3}} & \underline{\mathbf{6}} & \underline{\mathbf{9}} \\ \underline{\mathbf{6}} & \underline{\mathbf{14}} & \underline{\mathbf{20}} \\ \underline{\mathbf{9}} & \underline{\mathbf{20}} & \underline{\mathbf{29}} \end{bmatrix}$$

Verify: $\det(\mathbf{X}^\top \mathbf{X}) = \underline{\mathbf{0}}$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is singular and $(\mathbf{X}^\top \mathbf{X})^{-1}$ does not exist.

PRACTICE P3 · MULTICOLLINEARITY DETECTIVE

A data science team builds a model to predict monthly sales revenue using:

Feature	Description
X_1	Number of website visitors
X_2	Number of page views
X_3	Marketing spend (in $\times 1000$)
X_4	Number of social media ad clicks
X_5	Temperature in the city (Celsius)

- (a) Which pairs are *most likely* to exhibit multicollinearity?

Pair 1: X_1 and X_2

Reason: More visitors typically means more page views; they are highly correlated.

Pair 2: X_3 and X_4

Reason: Higher marketing spend directly drives more social media ad clicks.

- (b) If X_1 and X_2 have correlation 0.98, and $\beta_1 = 15.2$, $\beta_2 = -14.8$, what is wrong?

Coefficients are unstable and nearly cancel each other. Individually they are unreliable.

- (c) If the goal is only prediction, should they worry? [✓] No

Justify: For prediction only, multicollinearity is less harmful. The model can still predict well.

- (d) If $X_4 = 0.5 \times X_3$ exactly, what problem will OLS encounter?

Perfect multicollinearity: $X^T X$ is singular, $\det = 0$, OLS cannot compute $\hat{\beta}$.

SECTION 10 Closing Summary**▷ HOOK**

This lecture covered two major themes: how to extend Linear Regression to model curves, and how to verify that Linear Regression is trustworthy.

Part I — Polynomial Regression summary:

Concept	Key idea
Feature transformation	Create new columns (e.g. x^2) from existing features
Degree- n polynomial	$y = \beta_0 + \dots + \beta_n x^n$; each power is a new column
3D mental model	A curve in 2D becomes a plane in 3D feature space
Overfitting risk	High-degree polynomials can fit noise, not the real pattern

Part II — Five Assumptions summary:

Assumption	What it says	How to check	What to do if violated
Linearity	Linear relation between x and y	Scatter / residual plots	Transformations, poly terms
Normality of Residuals	Residuals $\sim N(0, \sigma^2)$	Histogram of residuals	Transform target, more data
Homoscedasticity	Constant residual spread	Residual vs $\hat{y}$ plot	Log / sqrt transformation
No Autocorrelation	Residuals are independent	Residual vs time plot	Add time features
No Multicollinearity	Features not highly correlated	Correlation matrix, VIF	Drop or combine features

PRACTICE P4 · COMPREHENSIVE CHALLENGE**(a) Match the Column.**

	Scenario	Action
1	Scatter plot of y vs x shows a clear U-shape.	(A) Check histogram of residuals
2	Two features have correlation > 0.95 .	(B) Apply log or sqrt transformation
3	Residual spread widens as $\hat{y}$ increases.	(C) Add polynomial terms
4	Team wants to know if residuals are bell-shaped.	(D) Remove one of the correlated features

Answers: 1 $\rightarrow$ **C** 2 $\rightarrow$ **D** 3 $\rightarrow$ **B** 4 $\rightarrow$ **A**

(b) True or False:

Statement	T/F	Justification
Polynomial Regression requires a different training algorithm than Linear Regression.	[F]	Same algorithm, different features
If R^2 increases when we add x^3 , the cubic term is always useful.	[F]	R^2 can increase for useless features
Heteroscedasticity means unequal error spread across prediction range.	[T]	Exactly the definition
Multicollinearity always makes predictions inaccurate.	[F]	Predictions can be fine; only coefficients are affected
Linearity requires every data point to lie exactly on a line.	[F]	Only the central trend must be linear

(c) Find the Mistake. “In perfect multicollinearity, $\mathbf{X}^\top \mathbf{X}$ has a determinant of 1, so it is invertible.”

The determinant is 0 (not 1). $\mathbf{X}^\top \mathbf{X}$ is singular and non-invertible.

- (d) **Predict the Outcome.** An ML engineer uses: square footage, number of rooms, and total area of all rooms (= sq. footage $\times$ fraction). What problem?

The third feature is a linear combination of the first two, causing multicollinearity. $\mathbf{X}^\top \mathbf{X}$ may be singular.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Explain why a straight line fails on curved data and how polynomial features fix it.
- ☐ Describe Polynomial Regression as a feature transformation, not a new algorithm.
- ☐ Visualize how a 2D curve becomes a plane in 3D feature space.
- ☐ Identify the risk of overfitting with high-degree polynomials.
- ☐ State and explain each of the five assumptions of Linear Regression.
- ☐ Check each assumption using scatter plots, residual plots, or histograms.
- ☐ Explain why perfect multicollinearity makes $(\mathbf{X}^\top \mathbf{X})^{-1}$ impossible.
- ☐ Distinguish when multicollinearity matters (inference) vs. when it does not (prediction).

Answer Key

- P1** (a) x^2 values: 4, 16, 36, 64, 100. (b) 3 columns (intercept, x , x^2). (c) Degree 1: 2 cols, no bend. Degree 2: 3 cols, yes (one bend). Degree 5: 6 cols, yes (up to 4 bends). Degree n : $n + 1$ cols, yes (up to $n - 1$ bends). (d) False. Same algorithm (OLS or GD); only features change. (e) (i) Both A and R are true, and R is the correct explanation of A.
- P2** (a) Linearity. (b) Homoscedasticity. (c) Normality. (d) Autocorrelation.
- P3** (a) Pair 1: X_1 & X_2 (visitors and page views). Pair 2: X_3 & X_4 (marketing spend and ad clicks). (b) Coefficients nearly cancel; individually unreliable. (c) No; for prediction only, multicollinearity is less harmful. (d) Perfect multicollinearity: $\mathbf{X}^\top \mathbf{X}$ is singular, OLS fails.
- P4** (a) $1 \rightarrow C$, $2 \rightarrow D$, $3 \rightarrow B$, $4 \rightarrow A$. (b) F, F, T, F, F. (c) The determinant is 0, not 1. (d) The third feature is a linear combination $\Rightarrow$ multicollinearity.